Variational and Optimization-based Approaches for Event-Based Vision Applications

From unobserved modes to well-posed estimators

Daniel Blauvelt

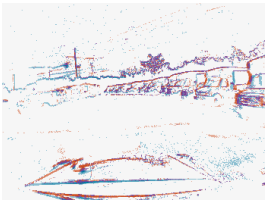
Department of Mathematics ▪ George Mason University
Dissertation Director: Harbir Antil

Dissertation Defense ▪ Fall 2026

What can changes tell us about a scene?



Reference image

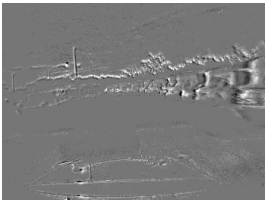


Accumulated events

50 ms sample

Blue +; orange −; purple: both.

→



Our reconstruction

What do the observations determine, and what does the model supply?

Takeaway. The observations constrain changes; the model supplies the remaining choices.

Dissertation: Chapter 2, saved MVSEC [35] outdoor_day1 reconstruction example.

The route through the talk

Sensor and data	What does an event camera report?
Reconstruction	How can changes recover an image over time?
Optical flow	How can local measurements determine motion?
Stereo depth	What changes when two cameras estimate depth?
Conclusions	What have we learned, and what comes next?

Reconstruction is the main focus, followed by the flow theorem and a brief stereo summary.

Takeaway. The common question: what do the observations determine, and what does the model supply?

Dissertation: Table 5.1.

Where this work fits

Image reconstruction

Rebecq et al. [12]: recurrent network, events to video.

Zhang, Yezzi & Gallego [6]: optical flow supplies a linear image inverse problem; classical or learned priors.

Antil–Sayre [2]: events and frames; bilevel estimation of an event-based blur-kernel parameter.

This work: per-pixel temporal inversion from event increments; kernel representations, activity-based penalties, and fixed-weight guarantees.

Optical flow

Horn–Schunck [16]: brightness constraint and spatial smoothness.

Benosman et al. [50]: timestamp geometry.

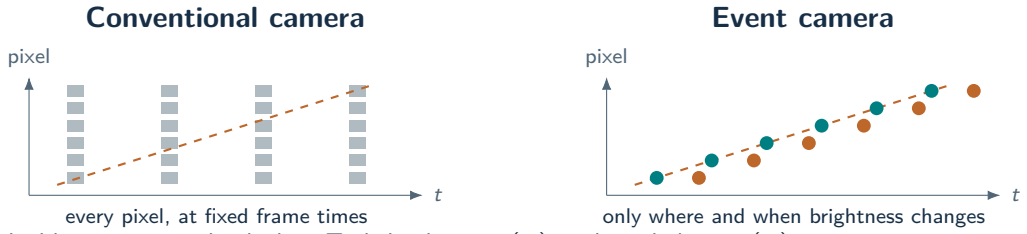
Weickert–Schnörr [57]: convex-regularizer framework and well-posedness theory.

This work: three event representations in one quadratic theory, with explicit identifiability and stability assumptions.

Takeaway. The contribution is the event-derived models and their precise guarantees: what the data determine and what regularization supplies.

Dissertation: Chapter 2 and Section 3.2; temporal publication: Antil, Blauvelt, and Sayre [17].

Why an event camera? Each pixel reports its own changes



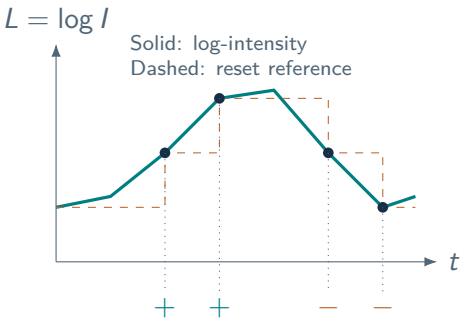
Dashed line: a moving bright bar. Teal: brightening (+); amber: darkening (−).

- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].
- Under steady illumination, motion makes events trace boundaries; illumination changes can also trigger events.

Takeaway. The sensor trades absolute brightness for precise timing of changes; each estimator supplies what the changes leave free.

Dissertation: Chapter 2, neuromorphic camera overview; Gallego et al. [1]. Diagram is schematic.

An event camera: each pixel reports threshold crossings



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

The preceding reset s_k^- is **local to that pixel**.

Takeaway. Events record changes. Adding a constant to a pixel's log-intensity leaves its events unchanged.

Dissertation: Eq. 2.1; Eq. 3.3; ideal threshold model [20]. Diagram is schematic.

From events to images, motion, and depth

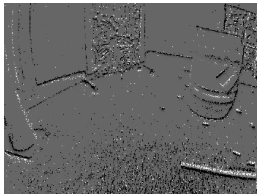
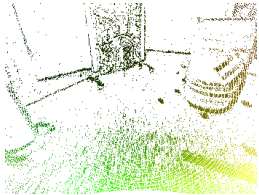


Image reconstruction

Ch. 2

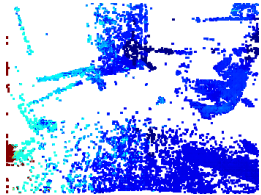
Grayscale: reconstructed value.



Optical-flow estimate

Ch. 3; event-active pixels

Color: direction and magnitude.



Stereo-depth estimate

Ch. 4

Depth: 0.5–8 m, blue to red (jet).

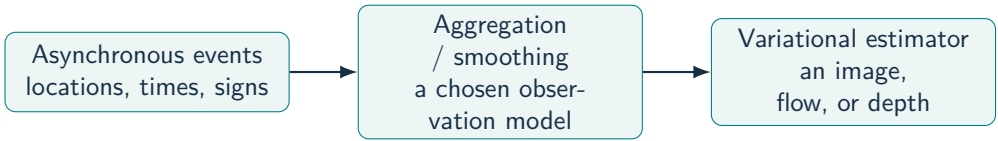
MVSEC indoor_flying2, near 47.6 s; stereo depth uses two cameras.

Takeaway. Each problem leaves a different choice: a brightness baseline, motion along an edge, and which pixels correspond.

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [35].

A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$



- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

Takeaway. The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Eq. 2.2–Eq. 2.4; Section 3.3; Section 4.3.3.

Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times t_i are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}.$$

$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

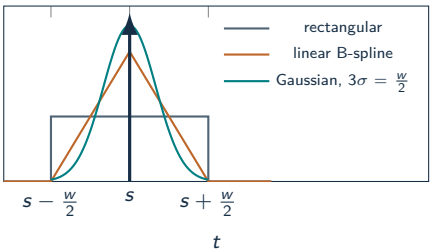
Events determine increments, leaving a constant free.

In scene terms: each pixel's log-brightness history is rebuilt on its own from how much it rose or fell in each interval. The model chooses the reference level; a pixel without events stays at that level, whatever its brightness.

Takeaway. Kernel width controls temporal averaging; a baseline requires an additional choice.

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

Temporal kernels: how far each event's influence spreads



Arrow: delta kernel (pure binning). Common support width w; schematic.

Each kernel has unit mass, is a translate, and has compact support:

$$\int_{\mathbb{R}} \phi_s = 1, \qquad \phi_s(t) = \phi_0(t - s).$$

Each event splits its sign p_k across the bins it overlaps:

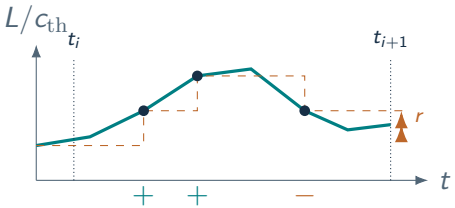
$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt, \quad \sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

Takeaway. Kernel width trades suppression of timing noise against blurring of rapid changes. The operator **A** is the same for every kernel.

Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

From threshold crossings to an approximate observation



Solid: L/c_{th}. Dashed: reference at the last event.

Ideal sensor: each event marks a log-brightness change of about one threshold c_{th}.

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt \;\approx\; \frac{L(t_{i+1}) - L(t_i)}{c_{\mathrm{th}}}.$$

Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}}.$$

Takeaway. Events approximate scaled brightness changes; the model supplies the level.

Dissertation: Physical connection (p. 24), Eq. 2.29, arbitrary-endpoint identity; threshold model [20]. Schematic diagram.

A regularized estimate is uniquely selected

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \left\{ \frac{1}{2} \|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \frac{1}{2} \|\mathbf{\Lambda}\mathbf{f}\|^2 \right\}.$$

The unique minimizer solves the normal equations:

$$\mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succ 0, \qquad \mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}.$$

Compute each pixel, then assemble frames

1. Form the pixel's matrix and right-hand side; solve the linear system numerically.
2. Run independent pixel solves in parallel; assemble their temporal profiles into frames.

The selected baseline satisfies $\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$: intervals with larger weights set the reference level.

Takeaway. The penalty chooses an unobserved baseline and also changes observable increments.

Dissertation: Eq. 2.8, Theorem 2.2.5; Tikhonov regularization [27]; monocular reconstruction implementation, per-pixel linear solves.

Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

One spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$: the bound limits how far one noise event moves a pixel's history.

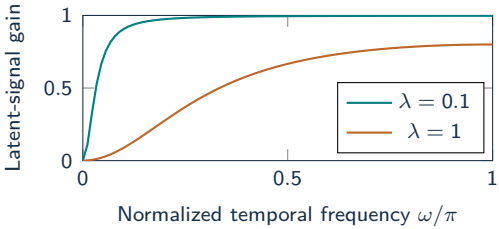
Takeaway. Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Theorem 2.2.5, Proposition 2.2.7.

What the penalty does to temporal information

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



The baseline is removed.

Low temporal modes are attenuated most.

A unique estimate does not restore the missing physical baseline.

In the images: slow brightness drift fades; passing edges remain.

Dissertation: Lemma 4.4.1, Theorem 4.6.21; cosine eigenbasis [81]. Curves: continuous extension.

How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here dc_i is the signed event accumulation; a_i is its magnitude, not a count of all events.

Rule	Sigmoid	Inverse proportional	Exponential
Reported outdoor CN	29.60	81.50	100.74

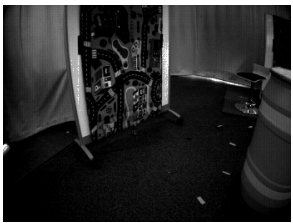
Selected by grid sweeps, balancing image fidelity and conditioning.
Quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.

Takeaway. Positive weights ensure a unique solve; the stability bound holds with weights fixed.

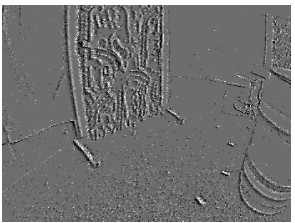
Dissertation: Adaptive temporal weights (p. 38); Table 2.7.

Temporal evidence: structure without an absolute baseline

MVSEC [35] indoor flying 2



Reference



Event-only reconstruction

MVSEC indoor aggregate results

Kernel	PSNR	CN
B-spline	8.86	22.16
Delta	7.70	1375.75
Gaussian	8.84	20.22

PSNR compares with a reference frame, so it also scores the absolute brightness that the model selects; the events constrain only changes.

Takeaway. Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Table 2.4; images: MVSEC indoor flying 2, Ch. 2 saved release.

Relation to Antil–Sayre bilevel reconstruction

This chapter
Events only $\longrightarrow$ activity-based $\Lambda(dc)$ $\longrightarrow$ fixed convex temporal solve.

Antil–Sayre [2]
Events and intensity frames $\longrightarrow$ outer optimization over z, a per-pixel, per-frame parameter of the event-based blur kernel.

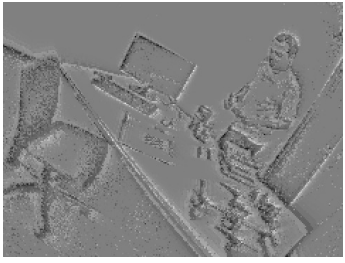
Sequence	Method	PSNR	CN
Indoor	Bilevel / event-only	15.37 / 10.85	39755 / 25.51
Outdoor	Bilevel / event-only	8.41 / 8.71	39755 / 27.10

PSNR: reference-image similarity (dB). CN: condition number.

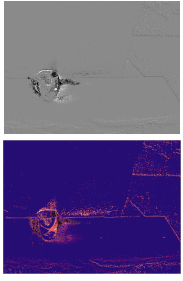
Takeaway. Frame information and a different optimization problem change the comparison.

Dissertation: Table 2.8 and Table 2.9. Antil–Sayre [2], Sec. 3, Eqs. (3.1)–(3.3).

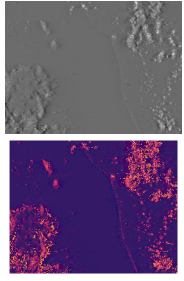
Real event data: indoor, high-speed, and orbital scenes



Office, handheld; ≈ 2.5M events



Bullet hits a cup; ≈ 350K events



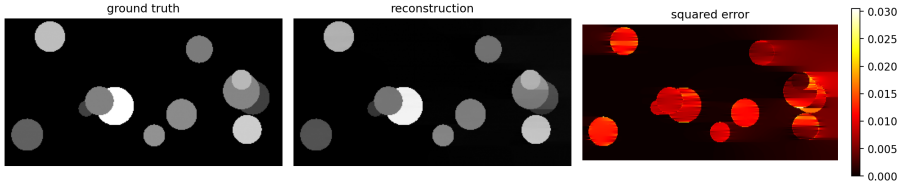
ISS (Falcon Neuro); ≈ 2M events

Cup and ISS: grayscale (top) and false color (bottom) of the same reconstructed frame.

Takeaway. One per-pixel fixed-weight solve, with sigmoid weights and no intensity frames, recovers structure across very different time scales.

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3. Data: office [36], cup [37], Falcon Neuro [38].

Shared motion couples temporal profiles to one scene



Simulated events (v2e [46]); the camera window shifts 1 px per view. Left: scene. Center: shared-scene estimate (smoothed-TV penalty). Right: squared error.

Chapter 2: each pixel has its own temporal profile.

Shared scene: known motion P maps one static scene θ into every view.

$$\underbrace{B_{\mathrm{SAR}}}_{\text{scene-to-event operator}} = (A \otimes I)\, P.$$

Takeaway. Known motion lets many views constrain one scene. Optical flow asks what happens when the motion is the unknown.

Dissertation: Figure 2.10, Eq. 2.47, Proposition 2.4.1; joint ESAR work [45]; synthetic aperture [98].

The aperture problem: what a local view reveals

An **aperture** is a restricted observation window: a small view of an edge [66].

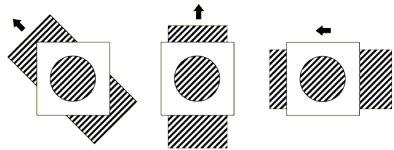
Through the windows



The visible stripe motions look the same.

Movie 5 source (GIF)

Reveal the full patterns



The patterns translate in different directions.

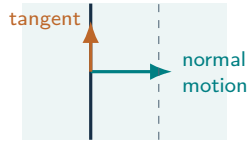
Movie 6 source (GIF)

Takeaway. An edge reveals perpendicular (normal) speed; motion along it (tangent) stays free. More observed directions can constrain the full velocity.

Perceptual illustration: Wolfe et al., *Sensation and Perception*, 4th ed., Sinauer © 2015. Activity 8.2, Movies 5–6.
Dissertation: Proposition 3.3.4 and the following aperture discussion.

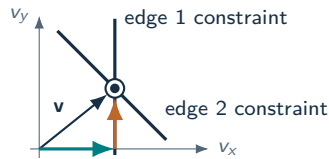
Motion introduces a new invisible direction

Image: one moving edge



A straight edge reveals its normal speed.

Velocity: what the events allow



Edge 1 fixes the normal part. If edge 2 shares the motion, it fixes the tangent part edge 1 left free.

The flow model in scene terms

- **Unknown:** image-plane velocity at every pixel.
- **Edges:** events give the speed perpendicular to each edge.
- **Neighbors move alike:** supplies motion along edges and where no events occur.
- **Requirement (main theorem):** edges in at least two orientations somewhere, so their lines cross.

Takeaway. As with the temporal baseline, the data leave one quantity free; smoothness carries information from other edge orientations to supply it.

Intersection of constraints [Adelson & Movshon 1982]. Dissertation: Proposition 3.3.4; requirement in Theorem 3.6.1.

Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

In scene terms: $|\nabla\mathcal{T}|$ is edge slowness, normal speed $1/|\nabla\mathcal{T}|$; motion along the edge leaves $\mathcal{T}$ fixed.

Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

Takeaway. Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Proposition 3.3.4; Definition 3.3.2; time surfaces [49, 50].

Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$; project over a window, $a(\mathbf{x}, s) = \int_s^{s+T} V(\mathbf{x}, t)\, dt$.

Modeling assumption (added to the threshold model)

The activity field moves with the scene: $V_t + \mathbf{v} \cdot \nabla V = 0$, with constant $\mathbf{v}$ over the window.

Integrating over the window then gives

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s),$$

so the voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

Takeaway. The threshold model supplies the activity field; transporting it with the scene is a separate modeling choice.

Dissertation: Section 3.4.1; derivative-operator qualifications; voxel grid [39].

The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

Takeaway. Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Section 3.4.2, Proposition 3.4.4; spatiotemporal-surface view [49].

Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

In scene terms: $\mathbf{B}$ records which motion directions the edges measure, and how strongly; b carries the measured values; the α term asks neighbors to move alike.

Takeaway. The representations change the coefficients; the analysis can be proved once.

Dissertation: Theorem 3.6.1 and its three corollaries; data-plus-smoothness form [16].

The main flow theorem: global information controls constants

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

The integrated matrix must identify **every constant velocity direction**.
In scene terms: edges in two or more orientations somewhere, so their constraint lines cross; neighbors supply the motion where $\mathbf{B} = 0$.

Takeaway. $\mathbf{B}$ may vanish on open regions; integrated directional information and smoothness still give a unique estimate.

Dissertation: Theorem 3.6.1, Eq. 3.59; classical quadratic-flow theory [57].

Proof idea: smoothness fixes variation, edges fix the drift

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

Smoothness controls variation

$$\|\mathbf{w}\|_{L^2} \le C_P \|\nabla \mathbf{v}\|_{L^2}$$

Poincaré [26]

Data control constant motion

$$m = \lambda_{\min}\Big(\int_\Omega \mathbf{B}\, d\mathbf{x}\Big) > 0$$

edges cover every direction

Together: coercivity

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) \ge c_0 \|\mathbf{v}\|_{H^1}^2$$

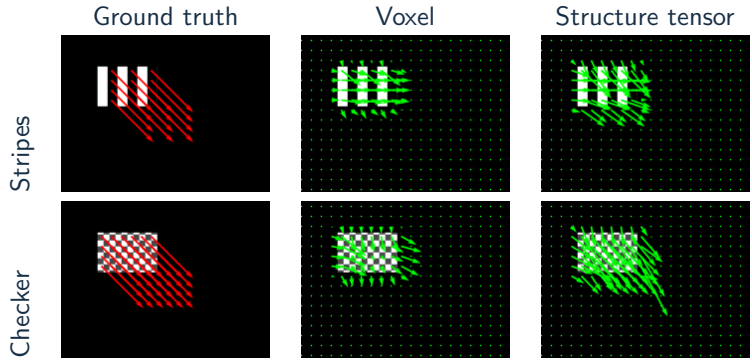
$$c_0 = c_0(\Omega, \alpha, \|\mathbf{B}\|_{L^\infty}, m)$$

In scene terms: smoothness leaves a camera pan free, so the edges must fix it; c_0 shrinks with m, as for nearly parallel stripes (next slide).

Takeaway. The gradient penalty controls nonconstant variation; integrated directional information controls the constant modes. Lax–Milgram [72] then gives the unique minimizer.

Dissertation: proof of Theorem 3.6.1.

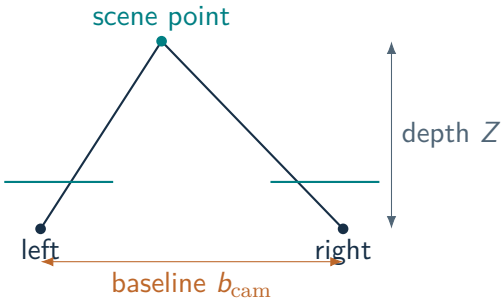
Known-motion examples make the aperture problem visible



Takeaway. Checker texture adds motion directions and reduces the aperture ambiguity; accuracy also depends on the representation and regularization.

Dissertation: Figure 3.4; v2e [46] simulation, same window and arrow scale.

Two viewpoints turn correspondence into depth



Rectification makes the correspondence search horizontal:

$$d = x_L - x_R.$$

$$Z = \frac{f_{\mathrm{cam}} b_{\mathrm{cam}}}{d}, \qquad d > 0.$$

Each candidate d pairs a left observation with $\mathbf{y}_R(x - d, y)$.

In scene terms: the true match sees the same scene point, so the two event profiles agree; each candidate is scored by how well one shared explanation fits both views.

Takeaway. The correspondence is unknown; solving the latent profiles at a fixed candidate is only the inner problem.

Dissertation: Section 4.5, Eq. 4.11; rectified stereo geometry [80]. Geometry is schematic.

Stereo on MVSEC: competitive accuracy, more scored points

At each candidate disparity d, solve two softly coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

Mathematical result (fixed d, frozen positive weights): Hessian $\succeq \lambda_{\min}^2 \mathbf{I}$; unique inner solve.
Benchmark: the full WLS method, with weight updates, spatial cost aggregation, and candidate selection.

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
Soft-coupled WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20
Xie adaptation [105]	93.43	78.60	93.66	66.57	59.36	1.39
Kogler et al. [104]	123.39	99.50	125.27	78.23	80.94	0.66
Rogister et al. [103]	142.17	128.20	153.32	114.80	90.32	1.01

MVSEC indoor flying 1–4, raw outputs. Each method is scored on its own estimates with ground truth in $[0.5, 8]$ m.

Takeaway. WLS: best on Flying 2 and log RMSE, within 6 cm elsewhere, over twice the points.

Dissertation: Theorem 4.6.3; Table A.5, Table A.4; MVSEC [35]. Xie and Piatkowska rows are local adaptations.

What the data constrain and what the models prove

Problem	What the observations constrain	What the mathematical model establishes
Temporal reconstruction	Represented brightness increments	Unique regularized history for each fixed positive weight matrix, with perturbation bounds
Optical flow	Event-derived directional motion constraints	Unique H^1 minimizer under integrated identifiability
Stereo depth	Paired temporal event-feature observations	Unique frozen-weight inner solve; the disparity comes from a separate comparison of candidates

Takeaway. Under the stated hypotheses, regularization makes each estimator determinate. Physical accuracy also depends on how well the representation describes the scene.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3; Table 5.1.

Research opportunities

Event-sensor modeling

Quantify the represented observation error: the kernel-boundary, overshoot, and sub-threshold terms of slide 11, plus timestamp estimation and binning. The stability bounds then translate a sensor-error bound into an estimate bound.

Learned temporal regularization

A possible extension: learn $\mathbf{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]. For $\mathbf{\Lambda} \succ 0$ the lower-level map $\mathbf{\Lambda} \mapsto \mathbf{f}^*(\mathbf{\Lambda})$ is single-valued and smooth.

Shared-scene and orbital sensing

Characterize the motion-invariant modes of B_{SAR} for a given trajectory (joint ESAR work [45]); apply to sparse orbital streams such as the ISS data.

Takeaway. Each direction starts from a quantity the dissertation controls: an observation error, a solution map, or a null space.

Dissertation: Chapter 5, Future Directions; physical connection (p. 24); Proposition 2.4.1. External paper cited above.

From event changes to well-posed estimators

Temporal: control an unobserved baseline.

Spatial: combine directional constraints globally.

Binocular: separate inner uniqueness from correspondence.

The common mathematical question is:

Which directions do the observations control, and which are selected by the model?

Thank you. Discussion and additional proofs follow.

Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. A gauge resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Better conditioning is not the same as smaller reconstruction bias.
- Reference-image scores do not identify an absolute intensity baseline from events.

Dissertation: Table 2.4, Table 2.5.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of a specified matrix	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. A stable estimate need not equal the physical signal.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate, not every ambiguity of the physical sensor.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

References (1 of 3)

[1] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, and Davide Scaramuzza. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1):154–180, 2022.

[2] Harbir Antil and David Sayre. Bilevel inverse problems in neuromorphic imaging. *Inverse Problems*, 39(9):094003, August 2023.

[6] Zelin Zhang, Anthony J. Yezzi, and Guillermo Gallego. Formulating event-based image reconstruction as a linear inverse problem with deep regularization using optical flow. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, 45(07):8372–8389, 2023.

[12] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. Events-to-video: Bringing modern computer vision to event cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3857–3866, 2019.

[16] Berthold KP Horn and Brian G Schunck. Determining optical flow. *Artificial Intelligence*, 17(1-3):185–203, 1981.

[17] Harbir Antil, Daniel Blauvelt, and David Sayre. Dynamic reconstruction from neuromorphic data. *Foundations of Data Science*, 2026. Early access. https://doi.org/10.3934/fods.2026017.

[20] Patrick Lichtsteiner, Christoph Posch, and Tobi Delbruck. A 128× 128 120 db 15 μs latency asynchronous temporal contrast vision sensor. *IEEE Journal of Solid-State Circuits*, 43(2):566–576, 2008.

[26] Lawrence C. Evans. *Partial differential equations*. American Mathematical Society, 2010.

[27] Andrey Nikolayevich Tikhonov and Vasiliy Yakovlevich Arsenin. *Solutions of Ill-Posed Problems*. V.H. Winston, 1977.

[35] Alex Zihao Zhu, Dinesh Thakur, Tolga Özaslan, Bernd Pfrommer, Vijay Kumar, and Kostas Daniilidis. The multivehicle stereo event camera dataset: An event camera dataset for 3D perception. *IEEE Robotics and Automation Letters*, 3(3):2032–2039, 2018.

References (2 of 3)

[36] Guillermo Gallego and Davide Scaramuzza. Accurate angular velocity estimation with an event camera. *IEEE Robotics and Automation Letters*, 2(2):632–639, 2017.

[37] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. High speed and high dynamic range video with an event camera. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(6):1964–1980, 2021.

[38] Matthew G. McHarg, Richard L. Balthazor, Brian J. McReynolds, David H. Howe, Colin J. Maloney, Daniel O'Keefe, Rayomand Bam, Gabriel Wilson, Paras Karki, Alexandre Marcireau, and Gregory Cohen. Falcon neuro: an event-based sensor on the international space station. *Optical Engineering*, 61(8):085105, 2022.

[39] Alex Zihao Zhu, Liangzhe Yuan, Kenneth Chaney, and Kostas Daniilidis. Unsupervised event-based learning of optical flow, depth, and egomotion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 989–997, 2019.

[40] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[45] Harbir Antil, Daniel Blauvelt, and David Sayre. ESAR: Event-based synthetic aperture reconstruction. arXiv:2607.15073 [eess.IV], 2026.

[46] Yuhuang Hu, Shih-Chii Liu, and Tobi Delbruck. v2e: From video frames to realistic DVS events. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1312–1321, 2021.

[49] Ryad Benosman, Sio-Hoi Ieng, Charles Clercq, Chiara Bartolozzi, and Mandyam Srinivasan. Asynchronous frameless event-based optical flow. *Neural Networks*, 27:32–37, 2012.

[50] Ryad Benosman, Charles Clercq, Xavier Lagorce, Sio-Hoi Ieng, and Chiara Bartolozzi. Event-based visual flow. *IEEE Transactions on Neural Networks and Learning Systems*, 25(2):407–417, 2014.

[57] Joachim Weickert and Christoph Schnörr. A Theoretical Framework for Convex Regularizers in PDE-Based Computation of Image Motion. *International Journal of Computer Vision*, 45(3):245–264, 2001.

References (3 of 3)

[66] David Marr and Shimon Ullman. Directional selectivity and its use in early visual processing. *Proceedings of the Royal Society of London. Series B. Biological Sciences*, 211(1183):151–180, 1981.

[72] Peter D Lax and Arthur N Milgram. Parabolic equations. In *Contributions to the Theory of Partial Differential Equations*, volume 33 of *Annals of Mathematics Studies*, pages 167–190. Princeton University Press, 1954.

[78] Ewa Piatkowska, Jurgen Kogler, Nabil Belbachir, and Margrit Gelautz. Improved Cooperative Stereo Matching for Dynamic Vision Sensors with Ground Truth Evaluation . In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 370–377. IEEE Computer Society, 2017.

[80] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2nd edition, 2003.

[81] Gilbert Strang. The discrete cosine transform. *SIAM review*, 41(1):135–147, 1999.

[98] Vaibhav Vaish, Gaurav Garg, Eino-Ville Talvala, Emilio Antunez, Bennett Wilburn, Mark Horowitz, and Marc Levoy. Synthetic aperture focusing using a shear-warp factorization of the viewing transform. In *2006 Conference on Computer Vision and Pattern Recognition Workshop (CVPRW'06)*, pages 129–129. IEEE, 2006.

[103] Paul Rogister, Ryad Benosman, Sio-Hoi Ieng, Patrick Lichtsteiner, and Tobi Delbruck. Asynchronous event-based binocular stereo matching. *IEEE Transactions on Neural Networks and Learning Systems*, 23(2):347–353, 2012.

[104] Jürgen Kogler, Christoph Sulzbachner, Martin Humenberger, and Florian Eibensteiner. Address-event based stereo vision with bio-inspired silicon retina imagers. In Asim Bhatti, editor, *Advances in Theory and Applications of Stereo Vision*, pages 165–188. InTech, 2011.

[105] Zhen Xie, Jianhua Zhang, and Pengfei Wang. Event-based stereo matching using semiglobal matching. *International Journal of Advanced Robotic Systems*, 15(1):1729881417752759, 2018.